AI, FINANCE, AND THE VIETNAM STOCK MARKET

Overview of AI in Finance

AI in finance includes: Machine Learning, Deep Learning, and Natural Language Processing (NLP).
Main applications: algorithmic trading, portfolio optimization, and risk management.
Global trend: AI in finance is expected to grow from **$10B (2023)** to **$42B (2030)**.
In Vietnam:

- **Opportunities**: 5% GDP growth, strong FDI, rapid digitalization.
- **Challenges**: inconsistent data, evolving regulations.

Current applications: price forecasting, sentiment analysis, fraud detection (e.g., Vietcombank).
Suggested learning resources: *Python for Finance* (Hilpisch), Coursera's *AI for Trading*, Vietstock API.
Conclusion: AI is no longer optional—it's essential for financial competitiveness.

Vietnam's stock market structure - How does Vietnam's stock market work?

Key exchanges:
- **HOSE** (Ho Chi Minh City Stock Exchange)
- **HNX** (Hanoi Stock Exchange)
- **UPCoM** (Unlisted Public Companies Market)

Main indices: **VN-Index** (broad market), **VN30** (top 30 large-cap stocks)
Securities available: Stocks, Bonds, ETFs
Market features:
- **Trading time** and **T+2 settlement rule**
- **±7% daily price limit**
- Focus on sectors: **banking, real estate, manufacturing**

Influencing factors:
- Macro policies, foreign investment (FDI), economic indicators

Investment methods:
- Technical Analysis (charts),
- Fundamental Analysis (financials),
- **Quantitative trading with AI**

AI techniques for financial applications - Which AI Tools are used in trading?

Supervised learning:
- **Regression**: predicts stock prices
- **Classification**: generates buy/sell signals

Unsupervised learning:
- **Clustering**: groups similar stocks
- **Dimensionality reduction**: simplifies complex financial data

Deep learning:
- **Neural networks & LSTMs**: great for time-series forecasting
- Capture complex patterns in stock data

Natural Language Processing (NLP):
- **Sentiment analysis** from news/social media
- Helps interpret market psychology

Reinforcement learning:
- AI agents learn to trade through trial and error
- Adapts to changing market conditions

Local adaptation:
- Works with **small datasets**
- Can process Vietnamese financial news for sentiment insights

Data for Vietnam stock market AI - How we prepared financial data

Data sources:
- **Vietstock**, **CafeF**, **Bloomberg**, **HOSE API**
- Social media, economic indicators (alt-data)

Data types:
- **Structured**: OHLC prices, volume, P/E, EPS
- **Unstructured**: news articles, reports, social posts

Preprocessing steps:
- Handle missing values, outliers
- Normalize and scale data
- Engineer features (e.g., MA, RSI)
- Prepare time-series format

Challenges in Vietnam:
- Limited historical data
- Language barriers in Vietnamese NLP
- Inconsistent data quality

Tools used:
- **Python (pandas)** for cleaning and transformation

How we built our trading strategy

Trading strategies covered:
- **Trend-following**, **mean-reversion**, **momentum**

AI in strategy design:
- **Logistic regression** and **Decision trees** classify buy/sell opportunities
- AI adapts to patterns in price and volume

Key trading components:
- **Entry/Exit rules** based on model output + technical indicators (e.g., MA crossover, RSI)
- **Stop-loss** to manage downside risk
- **Position sizing** to control exposure

Vietnam context:
- Strategy considers **T+2 settlement delay** and **±7% daily price limit**
- Simple models work well for noisy, low-liquidity stocks

Backtesting AI trading strategies

What is backtesting?
- Simulating a trading strategy using historical data
- Helps evaluate performance before live trading

Performance metrics used:
- **Sharpe ratio** – risk-adjusted return
- **Max drawdown** – largest loss from peak
- **Win rate** – % of profitable trades
- **Profit and Loss (PnL)** – total gains/losses

Backtesting steps:
- Load historical data
- Generate buy/sell signals
- Calculate daily returns
- Compute metrics and visualize

Vietnam-specific factors:
- **T+2 settlement**
- **±7% price movement limit**
- Data gaps due to illiquidity

Tools:
- **Python (pandas, NumPy)** for analysis
- **Matplotlib** for visualization

Portfolio optimization with AI

Goal: Balance **risk and return** by smartly allocating capital across assets
Modern Portfolio Theory (MPT):
- Developed by Harry Markowitz
- Maximize return for a given risk, or minimize risk for a target return

AI-enhanced optimization:
- Use **machine learning** to estimate expected returns
- Apply **reinforcement learning** for dynamic rebalancing

Vietnam-specific challenges:
- Fewer investment options
- High correlation among VN30 stocks
- Transaction costs and taxes matter more in smaller markets

Tools used:
- Python for mean-variance optimization
- Input: expected returns, covariance matrix
- Output: optimal asset weights

AI in Finance: Risk & Ethics

AI for risk management:
- **Value-at-Risk (VaR)**: estimates worst-case portfolio loss
- **Stress testing**: simulates extreme market conditions
- **Anomaly detection**: spots suspicious trades or data patterns
- **Fraud detection**: identifies unusual behavior in financial systems

Ethical considerations:
- **Bias in models**: results may be unfair if data is biased
- **Data privacy**: essential in financial environments
- **Compliance**: follow local laws (e.g., Vietnam's Decree 13/2023)

Risks in Vietnam's market:
- Market manipulation, insider trading, data leakage
- Cybersecurity is key to protect sensitive data

Tools:
- Python to calculate VaR and simulate risk
- Scikit-learn for anomaly detection



Project Components Overview

Data Collection
Gather and **preprocess financial data.**

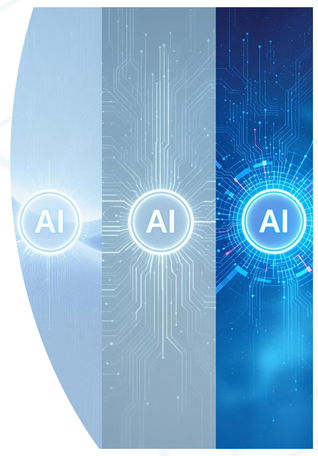
Model Selection
Choose predictive or optimization models.

Backtesting This Photo by Unknown Author is licensed under CC BY
Evaluate performance with historical data.

Data Collection and Preprocessing

Key Steps
- Identify reliable data sources.
- Cleanse and normalize raw data.
- **Feature engineering** for insights.

Why it Matters
Accurate data fuels intelligent AI models. Quality input ensures reliable output. This foundational step is critical.



AI Model Selection

Predictive Models
Forecast stock prices. Identify future trends.

Trading Models
Automate buy/sell decisions. Optimize execution.

Portfolio Optimization
Maximize returns. Minimize risk exposure. Balance assets.



Backtesting and Evaluation

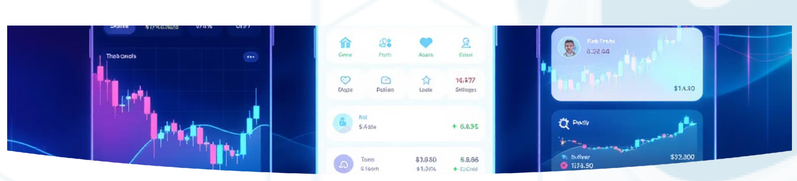
Evaluate your model's historical performance. Key metrics are essential for success.

1.5	20%	12%
Sharpe Ratio	**Max Drawdown**	**Annual Return**
Risk-adjusted return measure.	Largest historical loss.	Overall profit percentage.

Vietnam-Specific Considerations

Local Market Nuances
- Understand local regulations.
- Incorporate unique market dynamics.
- Adapt to regional trends.

Sector Selection
Focus on high-growth sectors. Banking, tech, and manufacturing are key. Local data sources are crucial for accuracy.



Product Development: MVP

Design a Minimum Viable Product. This ensures early user feedback. Focus on core functionalities.

1. **Essential Features**
   Basic trading functions. Portfolio visualization. Real-time data feed.
2. **Intuitive UI/UX**
   Easy navigation. Clear data presentation. User-friendly design.
3. **Scalability Plan**
   Design for future growth. Expand features gradually. Prepare for user increase.

Project Planning Essentials

Effective planning ensures project success. Define clear objectives and manage resources well.

Scope	Timeline	Deliverables
Clear project boundaries.	Set realistic deadlines.	Tangible outputs identified.
Define specific goals.	Allocate sufficient time.	Functional MVP.
Manage expectations.	Monitor progress.	Project proposal.

Tools for Success

Programming
Python is powerful. Utilize its extensive libraries. It's ideal for AI development.

Web Applications
Lovable AI for web apps. Create robust backend systems. Seamless integration is key.

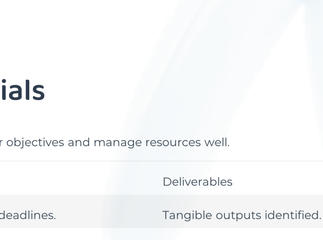
Visualization
Streamlit for quick dashboards. Build interactive UIs. Display data effectively.

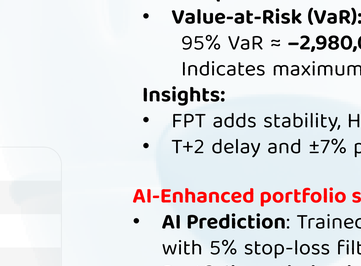
VIETNAM STOCK TRADING SYSTEM: STRATEGY BACKTESTING & OPTIMIZATION

- **Data**: HPG, FPT, VJC (daily stock prices)
- **Strategies**: MACD, RSI, Bollinger Bands, MA Crossover, Stochastic Oscillator
- **Best Strategies per Stock**:
  HPG: MACD (Sharpe: **0.381**)
  FPT: Stochastic (Sharpe: **0.732**)
  VJC: Stochastic (Sharpe: **0.012**)
- **Optimized Portfolio (Sharpe-maximizing)**:
  MACD (HPG): **34.7%**
  MA Crossover (FPT): **13.4%**
  Stochastic (FPT): **52.0%**
- **Performance**:
  Expected Return: **17.54%**
  Volatility: **19.03%**
  Sharpe Ratio: **0.921**
- **Insight**: Combining MACD & Stochastic strategies improves portfolio efficiency under Vietnam market constraints (T+2, ±7%).









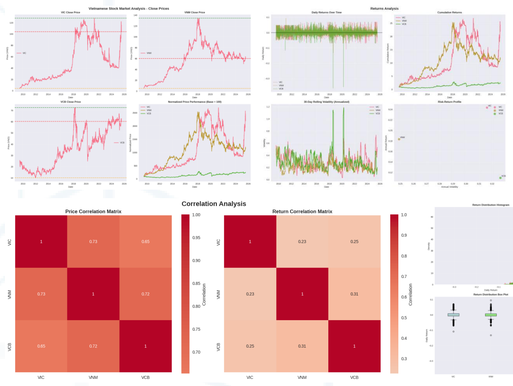
Data collection & Feature engineering

- ✓ We used the **VPS API** to fetch 600 days of data for three VN30 stocks: **VIC, VNM, VCB**.
- ✓ **Preprocessing on VIC**:
  - Moving Averages: 10-day & 30-day
  - Daily Price Change & Range
  - Lagged Volume
  - Target label: whether the price increases tomorrow
- ✓ The processed data is used to train AI models for stock prediction and portfolio strategy.
- ➢ Ready for smart AI-driven investment.

Date							
2024-01-22 07:00:00	43.30	43.350	43.651667	0.000000	2906906.0	0.35	0
2024-01-23 07:00:00	43.10	43.270	43.620000	-0.004619	1694115.0	0.50	1
2024-01-24 07:00:00	43.15	43.225	43.583333	0.001160	2213543.0	0.35	0
2024-01-25 07:00:00	42.80	43.140	43.546667	-0.008111	1695959.0	0.30	0
2024-01-26 07:00:00	42.80	43.120	43.513333	0.000000	1156705.0	0.35	0

Vietnamese Stock Analysis (VIC, VNM, VCB)

We analyzed 3 VN30 stocks using historical data from VPS API.
Key insights include:
- **Annual return**: 15–18%, with Sharpe Ratios > 1.0
- **Risk metrics**: Max Drawdown < 10%, daily VaR calculated
- **Daily & Cumulative returns**: Visualized for comparison
- **Volatility & Correlation**: Normality tested, rolling vol & heatmaps
- **Tools:** Python, pandas, matplotlib, seaborn
- ➢ **Result:** Strong, data-driven understanding of price movement, risk, and return to support smarter AI-driven investment strategies.



AI Model: Decision tree + Stop-Loss

- **Input features**:
  10-day & 30-day Moving Averages, Price Change, Lag Volume, Daily Range
- **Model**: Decision Tree (max depth = 5)
- **Train/Test split**: 70% train – 30% test
- ➢ **Stop-Loss rule**:
  If daily price drops ≥ 5%, predicted signal is forced to "no buy".
- ➢ **Model accuracy**: **44%**

Portfolio optimization with Risk controls

- **Objective**: Maximize Sharpe Ratio under constraints
- **Adjusted VIC return**: Based on AI signals from the decision tree
- **Constraints**: Max 40% per stock, weights sum to 100%
- **Optimal allocation**:
  VIC: 35.5%
  VNM: 40.0%
  VCB: 24.5%
- **Performance**:
  Expected return: 3613.79%
  Volatility: 330.68%
  Sharpe ratio: 10.92

Smart allocation + AI signals boost risk-adjusted return.

Risk analysis with Value-at-Risk (VaR)

- **Method**: Historical VaR
- **Confidence level**: 95%
- **Portfolio value**: 100,000,000 VND

Result:
- **95% VaR ≈ –3,049,146 VND**

→ *With 95% confidence, daily loss won't exceed ~3 million VND.*
A red dashed line on the histogram highlights this worst-case loss.

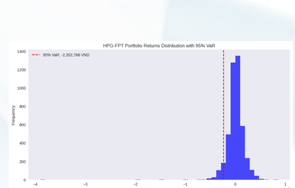


AI-Driven risk management on HPG & FPT

Task overview:
- Trained a decision tree to forecast **HPG** price direction.
- Integrated a **5% stop-loss** to reduce downside risk.
- Constructed an optimized portfolio with **HPG** and **FPT**, max 40% per stock.

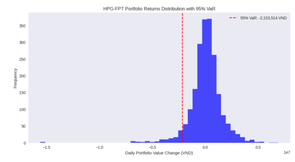
Results:
- **Accuracy**: 0.62 → Reliable model
- **Optimized weights**:
  HPG: 38.7%
  FPT: 61.3%
- **Sharpe ratio**: 8.43 → Strong risk-adjusted return
- **Value-at-Risk (VaR):**
  95% VaR ≈ **–2,980,000 VND**
  Indicates maximum daily loss (95% confidence) for a 100M VND portfolio.

Insights:
- FPT adds stability, HPG boosts return (but more volatile).
- T+2 delay and ±7% price limits in Vietnam may affect exit timing.



AI-Enhanced portfolio strategy with Risk controls

- **AI Prediction**: Trained Decision Tree for HPG price direction (Accuracy: **0.46**), with 5% stop-loss filter
- **Portfolio optimization**:
  Max weight per stock: **40%**
  Adjusted return using model signals
  Objective: Maximize **Sharpe ratio**
- **Optimal weights**:
  HPG: 40.00%
  FPT: 40.00%
- **Performance**:
  Return: 2,930.85%
  Sharpe ratio: 7.17
  95% VaR: -2.35M VND (on 100M portfolio)
- **Insight**:
  AI + risk filters improve strategy robustness under Vietnam market constraints (T+2, ±7%).



AI-Guided portfolio optimization with Risk controls

- **AI prediction**: Decision Tree on HPG (Accuracy: **0.46**) with 5% stop-loss
- **Optimized portfolio** (max 40% per stock):
  HPG: **32.05%**, FPT: **40.00%**, VJC: **27.95%**
- **Performance**:
  Return: **5,089.67%** Sharpe Ratio: **13.45**
  95% VaR: **–2.15M VND** (on 100M VND portfolio)
- **Insight**:
  AI signals + risk filters improve robustness under Vietnam's constraints (T+2, ±7%, API gaps)

